

Iris Flower Classification using Machine Learning

This project focuses on classifying Iris flowers into three species: Setosa, Versicolor, and Virginica using machine learning techniques. The classification is based on four features: sepal length, sepal width, petal length, and petal width.

Dataset Description

The Iris dataset is a well-known dataset used for classification tasks. It contains 150 samples with equal distribution among three species.

Methodology

1. Data loading from local CSV file
2. Exploratory Data Analysis (EDA)
3. Feature scaling using StandardScaler
4. Model training using Logistic Regression
5. Model evaluation using accuracy, classification report, and confusion matrix

Results

The Logistic Regression model achieved high accuracy (~95–100%) on the test data. Setosa species was perfectly classified, while Versicolor and Virginica showed minimal misclassification.

Conclusion

This project demonstrates the effectiveness of supervised machine learning algorithms in solving classification problems. The Iris Flower Classification model is simple, efficient, and suitable for beginners and internship-level projects.